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A Powered Prosthetic Intervention for Bilateral Transfemoral Amputees

Brian E. Lawson [Student Member, IEEE],

Department of Mechanical Engineering, Vanderbilt University, Nashville, TN, 37212 USA

Brian Ruhe,

College of Orthotics and Prosthetics, California State University Dominguez Hills, Long Beach, CA, 90822 USA

Amanda Shultz [Student Member, IEEE], and

Department of Mechanical Engineering, Vanderbilt University, Nashville, TN, 37212 USA

Michael Goldfarb [Member, IEEE]

Department of Mechanical Engineering, Vanderbilt University, Nashville, TN, 37212 USA

Abstract

This paper presents the design and validation of a control system for a pair of powered knee and ankle prostheses to be used as a prosthetic intervention for bilateral transfemoral amputees. The control system leverages communication between the prostheses for enhanced awareness and stability, along with power generation at the knee and ankle joints to better restore biomechanical functionality in level ground walking. The control methodology employed is a combination of an impedance-based framework for weight-bearing portions of gait and a trajectory-based approach for the non-weight-bearing portions. The control system was implemented on a pair of self-contained powered knee and ankle prostheses, and the ability of the prostheses and control approach to provide walking functionality was assessed in a set of experimental trials with a bilateral transfemoral amputee subject. Specifically, experimental data from these trials indicate that the powered prostheses and bilateral control architecture provide gait kinematics that reproduce healthy gait kinematics to a greater extent than the subject's daily-use passive prostheses.

Index Terms

Powered Prosthesis; Transfemoral; Robotics

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Competing interests: B.E.L., A.H.S., and M.G. hold patent applications through Vanderbilt University that have been licensed to Freedom Innovations, a United States-based prosthetics manufacturer.

I. INTRODUCTION

IT is difficult to estimate the number of bilateral lower limb amputees living in the United States. In the largest amputee demographic study known to the authors, persons suffering multiple amputations were not distinguished, although the total number of amputees was shown to be both large (with over 600,000 being either above the knee, below the knee, or the foot) and rapidly growing (projected to double over the next 50 years) [1]. The absence of an explicit report of the number of bilateral transfemoral amputees in the preceding study suggests that the population may be relatively small compared to the total number of persons living with a lower limb amputation.

Several studies do distinguish multiple amputations, however. For the period of April 1, 2006 through March 31, 2007, 4% of the 4574 referrals to the 43 prosthetics service centers in the United Kingdom were bilateral lower limb amputees [2]. In a review of 1846 lower limb amputees admitted to the Dundee Limb Fitting Centre in Scotland between 1965 and 1989, 18% were bilateral, with 3.6% being bilateral, transfemoral [3]. If these numbers are comparable for the U.S. population, then the number of persons living with bilateral, transfemoral amputations is likely to be quite small.

The size of the population could be a major contributing factor in the lack of dedicated research concerning prosthetic interventions for bilateral, transfemoral amputees. Such a small population does not draw a strong commercial interest in a free market society, and, as such, bilateral amputees that choose to utilize prostheses are forced to select from a family of prostheses developed almost exclusively for the needs of the unilateral amputee. A review that included 11 bilateral, transfemoral amputees from a particular regional amputee rehabilitation center over the period between July 1988 and December 1989 evaluated functional status at discharge from the center and 3 months later [4]. Of the 11 patients, none at any point were able to achieve either “limited community” or “community” ambulation classifications, and 6 of the 11 at the 3 month follow up were classified as “wheelchair-bound.” The “limited community” classification was defined as “a patient who can walk 152 m (500 ft), can participate in some avocational activity, but is not gainfully employed.”

A separate study investigating the mobility of wounded veterans included 50 bilateral lower limb amputees from both the Vietnam War and Operation Iraqi Freedom/Operation Enduring Freedom (OIF/OEF) [5]. 33% of the members of the Vietnam group abandoned their prostheses in favor of a wheelchair, while only 5% of the members of the OIF/OEF group did so. The vast majority of the abandoned prostheses in the Vietnam group were purely mechanical (as opposed to microprocessor-controlled passive devices), with a similar trend in abandonment for mechanical prostheses amongst the OIF/OEF veterans. This result suggests that increasing the functionality of the prosthesis may reduce the likelihood of abandonment, which has been shown to be quite high in bilateral, transfemoral amputees [6].

A. Bilateral Transfemoral Gait

Unilateral lower limb amputees typically exhibit significant compensatory behaviors in their intact limb (sound side) in order to accommodate the deficiencies of a typical passive

prosthesis. Such compensatory actions include a longer stance phase on the sound side (presumably for stability) [7], increased hip extensor and ankle plantarflexor work on the sound side [8], and vaulting (which is a mid-stance plantarflexion of the sound ankle in stance to provide extra ground clearance during swing of the prosthetic limb). These compensatory actions may be partially responsible for the increase in metabolic energy experienced by unilateral amputees during level walking [9–11].

Bilateral amputees have no intact limb to provide these compensatory behaviors, and they are significantly more limited in their ability to ambulate with prostheses. All control of and compensation for the prostheses must come solely from the hips. Since most prostheses are passive devices, the hips are over-taxed in terms of power generation, leading to excessive hip torques and a substantially increased metabolic cost of transport [12–15].

B. Emergence of Powered Prostheses

Traditionally, lower limb prostheses were passive by necessity; the technology required to construct a powered device comparable in size and weight to the anatomical limb has only emerged over the last five to ten years. Now that the supporting technology has arrived (power-dense motors/batteries and efficient microprocessors), the field of prosthetics is beginning to incorporate active elements. A number of research groups have published preliminary results on a variety of powered lower limb devices.

The prosthetics and orthotics company Ossur has released two versions of a powered knee joint called the Power Knee. This knee prosthesis is designed to be used in conjunction with an off-the-shelf passive prosthetic ankle/foot. Because the Power Knee has only existed as a commercial product, no literature is known to the authors regarding its design or control.

Herr *et al.* have developed a powered ankle for transtibial amputees which leverages both series- and parallel-elastic actuation [16–20]. By measuring the deflection of a series spring (with known spring constant K_s), an estimate of the ankle torque can be determined. An added benefit of series elasticity is that the introduction of compliance in between the prosthetic foot and the actuator softens the transfer of impulses from shock loading (i.e. heel strike), which may reduce wear on the mechanical transmission. The parallel spring reduces the peak torque required from the actuator at the high-power push off event in the late stance phase of level walking. In a study on 3 transtibial amputees comparing the use of a passive ankle with the powered prototype, it was shown that the powered ankle was able to reduce the energetic cost of transport by 7–20% [16]. In addition, the control approach demonstrated in [20] incorporates intrinsic speed adaptation, allowing an efficient gait across a range of walking speeds.

A separate effort in powered ankle design conducted by Sugar *et al.* has produced a two degree of freedom ankle named SPARKy 3, which stands for Spring Ankle with Regenerative Kinetics [21, 22]. An important goal in this effort was to enhance the agility of active users by providing the power necessary for running and jumping, along with providing actuation in the coronal plane. As a result, the designers opted to use two Maxon EC30 Powermax motors operating in parallel to provide 400 W of continuous power, with estimated peaks of up to 1500 W.

Lefeber *et al.* have just reported on another powered ankle that uses an electric motor to store energy in a spring during stance [23]. This design reduces the power requirements of the motor by spreading the energy delivery over a longer period of time (all of stance instead of the brief moment of powered push-off). Testing of the design shows energy delivery that matches or exceeds healthy subject data.

Other research efforts are using electromyography, inertial measurement, and other techniques to enhance control systems for powered knees or ankles [24–37]. To date, however, this is the first work known to the authors concerning the application of integrated powered knee and ankle prostheses for bilateral amputees.

II. METHODS

A. Powered Prosthesis

The authors have previously developed and reported on a powered transfemoral prosthesis prototype. The device has been shown to perform level walking [38], real-time intent recognition [39], upslope walking [40], ground adaptive standing [41], and stair ascent and descent [42]. Although the prototype (see Fig. 1) has undergone a series of revisions, the fundamental design has remained consistent. The joints are actuated by brushless DC motors rated nominally for 200 W continuous power (Maxon EC-4pole 30). They drive the joints through enclosed 3 stage transmissions consisting of a primary multi-V belt stage, a secondary roller chain stage, and a tertiary double-roller chain stage. The prototype can achieve approximately 100 Nm of torque at the ankle joint and 90 Nm of torque at the knee joint. A custom aluminum bridge connects the knee unit to the ankle unit and can achieve a minimum build height of 50 cm, as measured from the bottom of the prosthetic foot shell to the top of the pyramid connector. The structural components have been designed to accommodate an 85 kg user with a minimum safety factor of 2.

In the current design, the number of sensors on the prosthesis has been reduced by replacing the heel and toe load sensors with a single, axial load cell in the shank as a measure of weight bearing, and removing the optical encoders on the motors in favor of the integrated Hall Effect sensors as a measure of motor position and velocity. The joint angles are now directly measured with absolute magnetic encoders, which means that all control signals are derived from non-contact sensors, which is intended to substantially improve lifetime and reliability.

The prosthesis is controlled by an embedded system that consists of two microcontrollers: a 32-bit general purpose microcontroller for high level control and a 16-bit digital signal processor (DSP) for low level control. The 32-bit microcontroller runs the control system as described in this manuscript, and performs overhead functions such as data logging and inter-prosthesis communications. The DSP implements closed-loop current control on the knee and ankle motors and receives a current reference from the 32-bit microcontroller over a serial peripheral interface (SPI). The impedance control loop and state machines are updated at a sampling rate of 500 Hz, while the current control loop runs at approximately 5 kHz.

B. Walking Controller

The original control system for the unilateral transfemoral prosthesis consisted of a finite state-based impedance control framework, as described in [38]. For level walking, the gait sequence was parsed into discrete states accessed sequentially through biomechanical cues measured by sensors in the prosthesis. Within each state, a simple impedance control law was implemented to approximate the impedance seen in healthy gait. The following linear control law was utilized in each state:

$$\tau = k(\theta - \theta_{eq}) + b\dot{\theta} \quad (1)$$

where τ represents the commanded joint torque (for either the knee or ankle joint), θ represents the joint angular position, and $\dot{\theta}$ represents the joint angular velocity. k , b , and θ_{eq} are tunable parameters that define a virtual spring and damper, where k is the spring stiffness, θ_{eq} is the spring equilibrium position, and b is the damping coefficient.

In this paper, the control system includes provisions specific to bilateral applications. In particular, a data tether using the Controller Area Network (CAN) protocol has been implemented to share information between the two prostheses. At a rate of 500 Hz, the prostheses exchange the following signals:

- run_time – counter that increments every sample
- control_modes – state information for the control system
- knee_position – angle of the knee joint
- knee_velocity – velocity of the knee joint
- ankle_position – angle of the ankle joint
- ankle_velocity – velocity of the ankle joint
- shank_load – axial load in the shank
- prc_stride – percentage of stride for the current stride

The number of states and state behavior has also been modified from the original finite state-based impedance control framework. The bilateral controller has three distinct states for the level walking gait cycle (see Fig. 2). In each state, eq. 1 is implemented for both the knee and ankle joints. In the stance state, the impedance parameters are constant, thereby producing a locally passive system that will always come to rest in the absence of disturbances or input from the user. In this state both joints are tuned to behave as stiff, over-damped systems when supporting the user's body weight. Neither joint should be overly stiff, however, to allow for stance knee flexion and ankle plantarflexion for shock absorption after heel strike.

In the push-off state, the ankle impedance parameters are also constant, though the equilibrium position of the ankle is moved from a neutral value (tuned specifically for the user) to an appropriate plantarflexed value to generate the energy and motion for powered push-off. By arbitrarily changing this equilibrium value, a finite amount of energy is introduced to the system, creating a transient response but then immediately reverting to a

passive system within this state. Push-off cannot be repeatedly excited due to constraints on allowed state transitions such as the requirement for alternating strides described subsequently.

In order to simplify the tuning process, however, the knee behavior is modified in the push-off state. While the stiffness and damping coefficient of the knee joint remain fixed during push off, the equilibrium position becomes a function of the ankle angle. During push off, which takes place between approximately 45% and 60% of stride (see Fig. 3), the knee angle and ankle angle have an approximately linear correlation. This observation has been noted previously in the literature and leveraged for the design of a passive prosthesis [44]. A linear least squares relationship was determined from the healthy data presented in [43] and is depicted in Fig. 4. This relationship is used to continuously calculate the knee equilibrium position in the push off state. As a result, the knee and ankle angles are kinematically linked in this state. As the user unloads the ankle, the knee begins to yield in a synchronous manner, giving the user direct control over the duration of this state through the modulation of load on the prosthesis.

The third and final state consists of the swing phase of gait. Because the prosthesis is non-load bearing in this state, the classical advantages of pure impedance-based control would not be fully realized. The two primary advantages of the finite state-based impedance control framework presented in [38] are that the behavior of the prosthesis is not explicitly time-dependent (i.e. the user, as opposed to the device, has control over the speed at which the device moves), and that the forces produced by the prosthesis are specifically tuned to interact safely with the environment. These features are in direct opposition to the nature of high-gain trajectory control, which specifies precisely the time-dependency of the motion at a high, disturbance-rejecting impedance. If the walking speed is known, however, the time-dependent nature of trajectory execution in the prosthesis simplifies to a problem of proper initiation of the trajectory. This is the same problem seen in the pure finite state-based framework, as the thresholds for state transitions must be timed accordingly for each user. Additionally, the high impedance used in trajectory tracking will not accommodate unknown interactions in the environment, such as the varying load placed upon the prosthesis by the user. In a non-weight bearing phase of gait such as swing, however, external loads are minimal. Since the authors have previously demonstrated an algorithm to detect and classify stumbles [45], a robotic prosthesis could easily be programmed to change its behavior in the event of a stumble, meanwhile retaining high gain trajectory control during the swing phase.

For these reasons, the swing phase of the walking controller has been reduced to a single state that executes appropriate trajectories at the knee and ankle joints. Upon completion of the trajectories, the controller reverts to the initial state (stance), previously described. The trajectories are produced by a spline interpolation generated upon entry into the state. Both the shape and duration of the trajectories are determined by an estimate of the cadence, as subsequently described.

C. Inter-prosthesis Communication

Using the control signals exchanged between the prostheses, each prosthesis implements several safety features to avoid inappropriate mode transitions which could potentially destabilize or injure the user. Firstly, the prostheses are constrained to only allow alternating steps. Neither prosthesis can enter the push off state a second time before the contralateral prosthesis has done so. There is a 2 second time-out on this feature, such that standing for a moment will re-enable push off on both prostheses. Additionally, once one prosthesis has entered the push off state, the contralateral prosthesis is locked out from also entering the push off state until the ipsilateral prosthesis has reached the last 10% of the stride. It is important to allow the triggering of a contralateral push off before ipsilateral heel strike in order to achieve dynamic walking similar to healthy biomechanics. An earlier push off, however, could result in premature destabilization on the ipsilateral (stance) leg and might produce a fall, and, as such, must be avoided.

D. Variable Cadence

The stance and push off states of the controller inherently accommodate a variable cadence, since there is no explicit time dependence in the control law. In these phases, therefore, the user can control cadence simply by modulating the speed at which he or she moves the hip. At faster cadences the ankle transition (State 0 to State 1) and the load transition (State 1 to State 2) occur sooner. For the swing state, which executes trajectories for the knee and ankle joints, the duration and shape of the trajectories are altered for each stride. Holgate *et al.* have presented a similar approach for a powered ankle in [46]. A timer is initialized at the beginning of the push off phase (Phase 1), and its value at the end of this phase determines the speed and shape of the trajectory executed in the swing phase (Phase 2). At this point in the stride, the speed of the stride is determined until the subsequent heel strike, at which point the control behavior reverts to the non-time-varying impedance control law of eq. 1. Although the duration of the push off state is only a portion of the stance timing available for cadence estimation, allowing the beginning portion of stance to be independent of cadence provides an important safety and control mechanism for bilateral amputees. During the stance phase (early and middle stance, specifically), a bilateral amputee may take more time than a healthy subject to stabilize at the hip and achieve an appropriate balance. This time may not be a function of the user's desired cadence, and, consequently, it is not used in the cadence estimation.

The duration of the trajectories executed in the swing phase are directly proportional to the duration of the previous push off phase. The shape of the trajectories, however, is also adjusted according to this measure of cadence. Three reference trajectories are specified at nominal, slow, and fast cadences via seven points. The duration of time spent in the push off phase is used to interpolate between the closest two of these three sets of points, creating a new spline that corresponds to the cadence. Although the differences between the reference splines are adjustable at each control point, in this work the only control points that were modified were the ones that affected the maximum knee flexion, as can be seen in Fig. 5. Effectively, the maximum knee flexion is reduced at slower cadences and increased at faster cadences.

E. Experimental Validation

The control system was implemented on a pair of prostheses and vetted through the use of able body adapters on healthy subjects [47]. The system was then implemented on a bilateral transfemoral amputee subject (one of the authors). The subject was 38 years of age, and his bilateral amputations were the result of a traumatic injury at age 18. He wore Ossur Rheo knees (microprocessor controlled knees, or MPCs) and College Park Soleus ankles (dynamic-elastic response ankles, or DERs) for daily use and for comparison in this study. He weighed approximately 70 kg at the time of the testing. After several testing and development sessions, a kinematic assessment was performed on both daily use (passive) prostheses and the powered prostheses. Approval for this study was given by the Vanderbilt University Institutional Review Board and informed consent was obtained before the assessment.

For each pair of prostheses, the subject walked at two comfortable walking speeds of his choosing over ground in a motion capture laboratory. He was fitted with a full skeletal marker set for motion capture with a NaturalPoint Optitrack motion capture system. Data was collected for ten trials using both the powered prostheses and the subject's daily use, passive prostheses. Because of the limited capture volume (approximately 36 square meters), the middle two strides were selected from each trial for analysis. The motion capture data was exported in the Biovision Hierarchy (BVH) file format and imported into Matlab for post-processing. Heel strikes were manually selected in Matlab and the Cardan angles specified in the BVH file were converted to homogeneous transformation matrices. From these matrices, sagittal joint angles were extracted for the hip, knee, and ankle joints for each limb. Each stride was then normalized in terms of percentage of stride using the heel strikes selected previously.

III. RESULTS

The joint angles for the right side hip, knee, and ankle are plotted in Fig. 6 for able bodied subjects (reprinted from [43]), the amputee subject using the powered prostheses, and the amputee subject using his daily use, passive prostheses. A video is included in the supplementary that shows the subject walking on both pairs of prostheses outside of the motion capture laboratory. The healthy subject data did not explicitly state walking speeds or cadences, but the data reprinted here are specified as a slow cadence, which most closely represents the subject's selected cadences. On the powered prostheses the subject walked with an average cadence of 66 steps/min (step length of 1.23 m and speed of 0.68 m/s) at a self-selected slow speed and 83 steps/min (step length of 1.44 m and speed of 0.99 m/s) at a self-selected normal cadence. On his daily-use prostheses the subject walking with an average cadence of 70 steps/min (step length of 1.05 m and speed of 0.61 m/s) at a self-selected slow speed and 97 steps/min (step length of 1.31 m and speed of 1.06 m/s) at a self-selected normal cadence. The subject used a stability aid in his right hand for all trials (a forearm crutch for the powered trials and his daily use cane for the passive trials). Several representative photographs of the subject walking with the powered prostheses are shown in Fig. 7.

In addition to the motion capture data, the powered prostheses log internal data at a rate of 250 Hz, providing an estimate of joint torques through the measurement of the motor currents. An approximation of the amount of mechanical energy delivered can be made using the motor current, motor torque constant (0.0276 Nm/A), the drive train's transmission ratio (196:1), and the ankle velocity. This approximation can be refined with an empirical measure of the transmission's frictional characteristics. In this case, the friction in the transmission was modeled as a combination of Coulomb friction and viscous damping. The model parameters were estimated by iteratively increasing their values from zero until the system no longer remained stable with otherwise zero input. With this approach, the system demonstrated stability with Coulomb friction no higher than 5 Nm and a damping coefficient no higher than 0.01 Nm·s/deg. Fig. 8 depicts the powered ankle behavior for a characteristic stride (at the normal self-selected cadence), including the torque and power estimates. An integration of the power in the stance and push off phases yields -7.67 J and 11.47 J, respectively, demonstrating that this behavior would not be possible in a passive system and confirming that the powered prostheses delivered powered push-off.

IV. DISCUSSION

The powered joint data show several kinematic features of healthy gait that are absent in the passive data. Most notable, perhaps, is the evidence of powered push-off from the ankle joint. The powered ankle exhibits evidence of an active push-off with a strong plantarflexion of the joint just after 60% of stride. This phenomenon can be verified by consulting the power plot in Fig. 8. During the push off phase, the powered ankle delivers almost 50% more energy than all of the energy both dissipated during heel strike or (virtually) stored during middle stance. Contrarily, the passive ankle returns to its nominal position (zero degrees) once unloaded, and, although its energy contribution is not explicitly measured, it is fundamentally constrained to return only an amount of energy equal to or less than what was stored through dorsiflexion in middle stance.

For the prosthetic knee joint, a period of stance knee flexion is observed in the powered trials. Stance knee flexion aids in shock absorption from heel strike and helps to lower the body's center of mass through segments of the stride. The MPC knees cannot provide an extensive torque greater than the externally applied torque since they only exhibit passive behavior. As a result, stance knee flexion cannot be achieved with such a device unless the user supplies a torque from the hip to extend the knee once it flexes. Such an action is not biomimetic, and it also gives the user a sense of instability in stance. As such, the passive data show a strongly hyperextended knee in stance that breaks (moves past zero into flexion) only at the initiation of swing. The ability of the powered prostheses to provide full support with flexed knees is a feature that could also enhance safety for bilateral amputees, as it is safe for the user to load the prosthesis regardless of the knee configuration. Specifically, the powered prostheses will provide supportive torque at all times in this controller, therefore eliminating the stance phase yielding that would occur in an MPC knee that was loaded whilst not in hyperextension. Furthermore, the active swing from the powered prosthesis is guaranteed (within the torque limits of the device) to extend and provide support, even in the event that it experiences a disturbance such as a stumble or scuff. Finally, landing

prematurely on the device will not sacrifice its ability to support the user, as the active swing extension will still extend the knee.

Although the subject's cadences were slightly slower with the powered prostheses at both speeds, the step lengths were increased. As a result, the overground walking speeds were comparable for both cases. The increased step length is partially due to the fact that the minimum build height of the powered prosthesis prototypes was slightly too tall to match the build height of the passive prostheses. The subject's overall leg length was increased by approximately 7 cm (6.7%) when using the powered prostheses. The step lengths increased by 17% and 10% for slow and normal cadences, respectively, so it is possible that the powered prostheses compelled the subject to further increase the step length independent of their longer length.

For both the powered trials and the passive trials, the intact hip joints exhibited comparable ranges of motion which were significantly larger than that seen in healthy subjects. Overactive hip motion is consistent with the literature that points to increased hip torques and metabolic costs for bilateral transfemoral amputee gait [12–15]. It is anticipated by the authors, however, that the restoration of active behavior at the ankle will serve to alleviate this over-taxing of the hips in the long term. However, this particular subject has approximately 20 years of training on passive devices and only several hours of development and testing on the powered prosthesis prototypes. As such, it is not surprising that learned kinematic compensations at the hip were not readily modified by the restoration of powered ankle push-off. Despite this fact, these data show that a pair of powered knee and ankle prostheses can be designed and controlled such that a bilateral transfemoral amputee can achieve a gait that better reflects healthy kinematics at the knee and ankle joints.

V. CONCLUSION

The design and control of a powered prosthetic intervention for bilateral transfemoral amputees has been presented and demonstrated on an amputee subject. The control system can achieve variable cadence walking with a reduced number of user-dependent parameters relative to previously reported controllers for powered transfemoral prostheses. The prosthesis prototypes and control system described in this work have been shown to provide improved kinematics for level walking at the prosthetic knee and ankle joints over the subject's daily use prostheses, relative to healthy data. The authors plan to extend this controller to other lower limb activities, as well as investigate lower limb kinetics in bilateral transfemoral amputees in future work.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Biographies



Brian Lawson (S'09) received the B.A. degree in physics from Grinnell College, Grinnell, IA, in 2006. He received the B.S. degree in mechanical engineering from the Fu Foundation School of Engineering and Applied Science, Columbia University, New York City, NY, USA, in 2007, and the M.S. and Ph.D. degrees in mechanical engineering from Vanderbilt University, Nashville, TN, USA, in 2011 and 2014, respectively. He is currently a postdoctoral scholar at Vanderbilt University, Nashville, TN. His research interests include robotics, mechatronics, and applied controls.



Brian Ruhe received the B.S. degree in biomedical engineering from Wright State University in 1998, and the M.S. and Ph.D. degrees in biomedical engineering from Northwestern University in 2004 and 2009, respectively. In 2012, he received the American Board of Certification's certification of prosthetics. Currently he is an assistant professor at California State University, Dominguez Hills. His current research interests include rehabilitation robotics and human-prosthesis interface technology.



Amanda Shultz (S'11) received the B.S. degree in mechanical engineering from the Ogden College of Science and Engineering at Western Kentucky University, Bowling Green, KY, USA, in 2011. She is currently a Graduate Research Assistant at Vanderbilt University, Nashville, TN, USA. Her research interests include design and control of electromechanical systems.



Michael Goldfarb (S'93 – M'95) received the B.S. degree in mechanical engineering from the University of Arizona, Tucson, AZ, USA, in 1988, and the S.M. and Ph.D. degrees in mechanical engineering from Massachusetts Institute of Technology, Cambridge, MA, USA, in 1992 and 1994, respectively. Since 1994, he has been with the Department of Mechanical Engineering, Vanderbilt University, Nashville, TN, USA, where he is currently the H. Fort Flowers Professor of Mechanical Engineering. His current research interests include the design and control of advanced upper and lower extremity prostheses and gait restoration for spinal cord injured persons.

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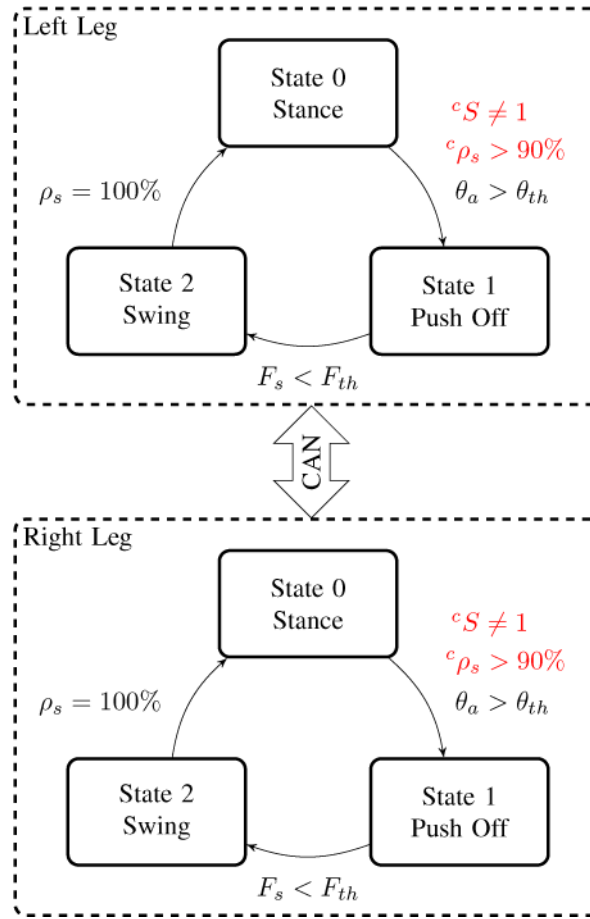
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Fig. 1.
The pair of powered prostheses used in the experiments.

**Fig. 2.**

The finite state machines executed by the prostheses. θ_a is the ankle joint angle, which is compared to a predetermined threshold, θ_{th} , to trigger the transition into the ankle push off state. F_s is the axial load in the prosthetic shank, which is compared to a predetermined threshold, F_{th} , to trigger the transition into swing. The swing state executes a trajectory and automatically reverts to the stance state when the trajectory ends, which corresponds to the percentage of stride, ρ_s , reaching 100%. The red conditions on the transition from State 0 to State 1 are safety conditions that are dependent upon the contralateral signals ${}^c\rho_s$ and cS , which are the contralateral percentage of stride and contralateral state, respectively.

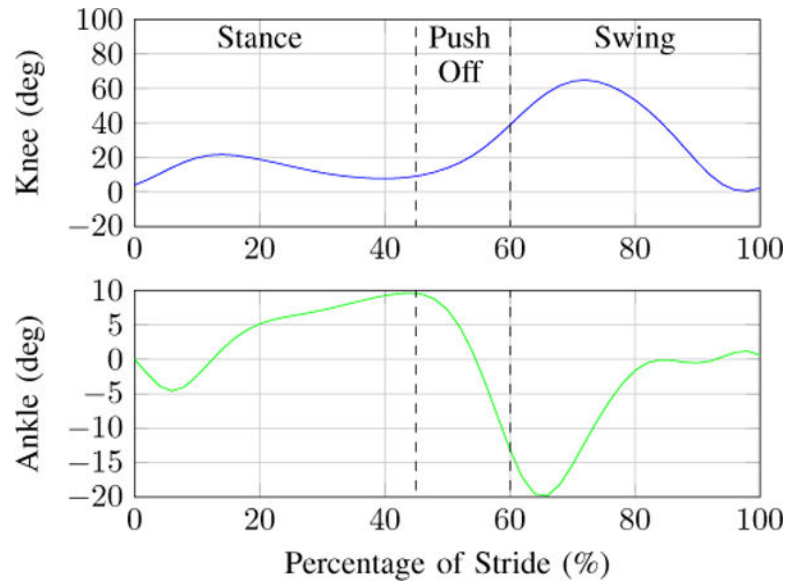
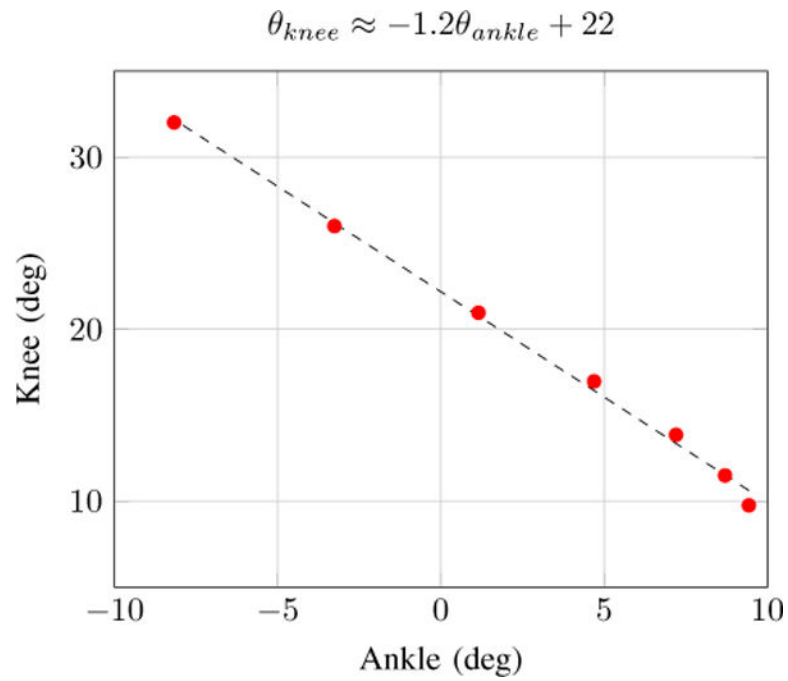


Fig. 3.
Healthy subject joint behavior reprinted from [43].

**Fig. 4.**

The approximately linear relationship between knee angle and ankle angle during level walking in healthy subjects between 45% and 60% of stride.

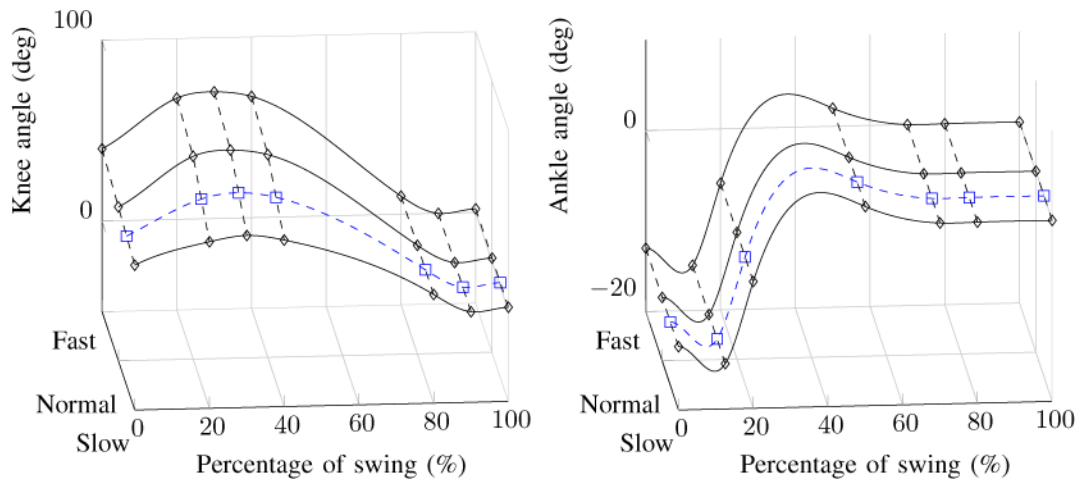
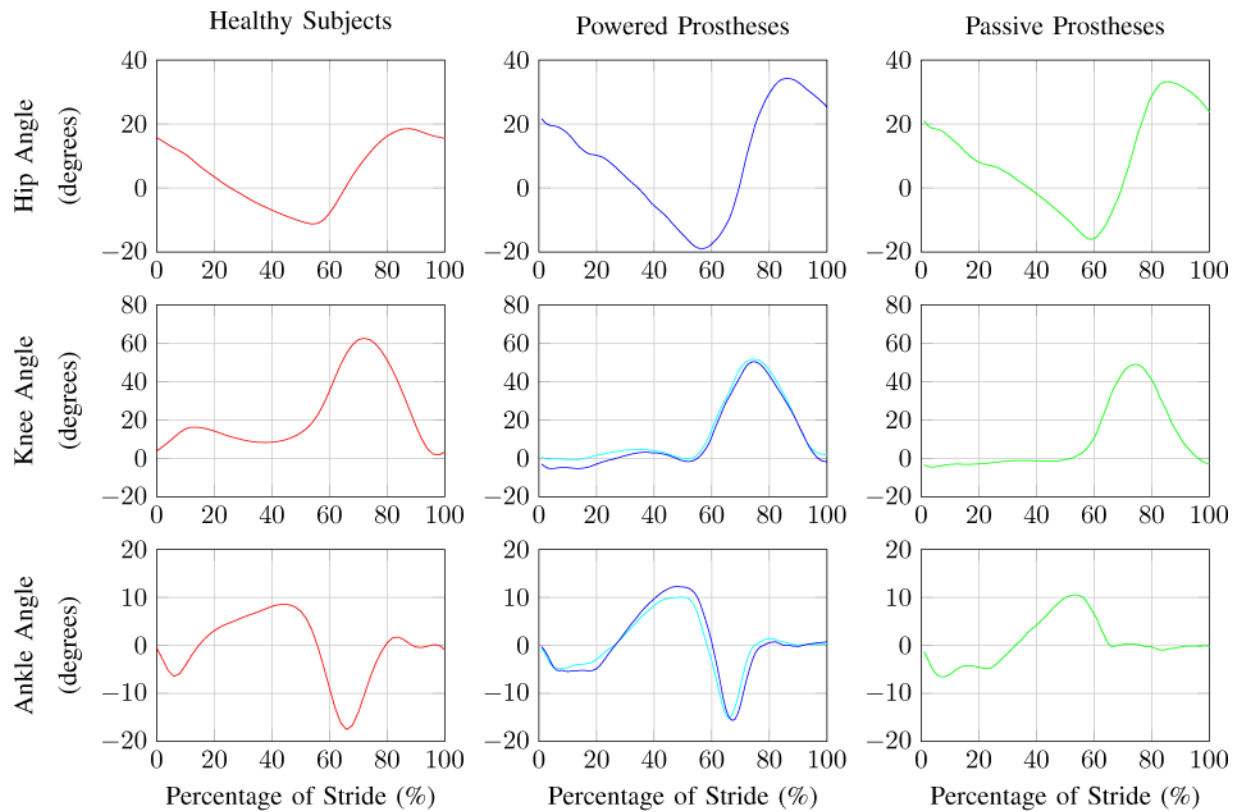


Fig. 5.

The three reference trajectories and an estimated trajectory for the swing phase of walking at the knee and ankle joints.

**Fig. 6.**

Joint angles for the right side hip, knee, and ankle for able bodied subjects (reprinted from [43]), the amputee subject using the powered prostheses, and the amputee subject using his daily use, passive prostheses. The knee and ankle joint plots for the powered prostheses include the internally measured joint angles from the powered prosthesis in lighter blue.



Fig. 7.
The subject walking with the powered prostheses.

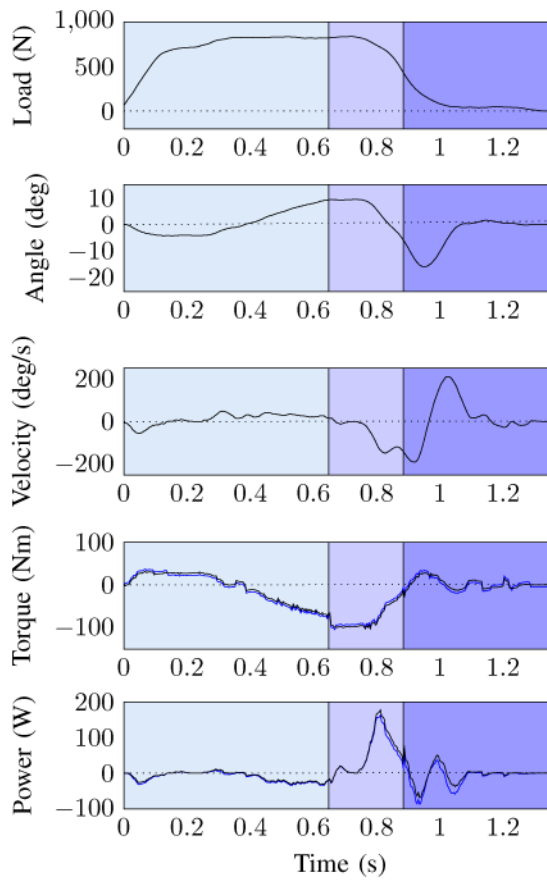


Fig. 8.

The powered ankle behavior for a characteristic stride. The top plot shows the axial load in the shank as measured by the onboard magnetic load cell. This signal is roughly proportional to the vertical component of the ground reaction force. The next two plots show the ankle angle and angular velocity, respectively. In the fourth plot, the black line represents the output torque of the ankle given 100% transmission efficiency, and the blue line represents the output torque assuming the transmission friction model described. The final plot shows the estimated power output of the ankle without (black line) and with (blue line) the transmission friction model. The three colored regions of the plots denote the three phases of the controller: stance, push off, and swing. With the transmission friction model, the net energy contribution in the stance phase was -7.67 J, while the net energy contribution in the push off phase was 11.47 J.